PHYSICS + HONORS PHYSICS / KINEMATICS / BUILD YOUR OWN REFERENCE

Unit 1 Cheat-Sheet Toolkit

Build a card that helps you choose your next step. **The building is part of the studying.**

Use the class card rules: one side of one 4 × 6 index card, handwritten by you. Equations, definitions, triggers, and reminders only; no worked problems. Follow any updated exam-specific directions from Mr. Avendano. This one-pager is a planning tool.

1 / Choose what earns space

1. **Recall first.** List the Unit 1 ideas you remember without notes.
2. **Check your corrections.** Star repeated mistakes on PS4-6 and the review.
3. **Compress.** Turn each mistake into a short cue you can explain.
4. **Draft, then handwrite.** Use clear headings and readable symbols.

Useful cue: "Same time; separate x/y."
Skip: paragraphs, copied solutions, facts you know cold.

2 / Give the card four zones

GRAPH RULES Slope, area, signs, units	**EQUATION CHOOSER** Formula + when it helps
FREE FALL / PROJECTILES Axes, components, shared time	**MY ERROR CHECKS** Your most useful reminders

Add a narrow bottom strip: **K → D → E → A**.
Givens + units → target → equation → rearrange, substitute, check.

Put your biggest trouble spot first. Avoid tiny writing.

3 / Add cues, not just formulas

Constant acceleration in one axis: choose the equation that leaves out the variable you do not have or need.

EQUATION	LEAVES OUT
$v_f = v_i + at$	Δx
$\Delta x = v_i t + \frac{1}{2}at^2$	v_f
$v_f^2 = v_i^2 + 2a\Delta x$	t
$\Delta x = \frac{1}{2}(v_i + v_f)t$	a

Vertical motion: use Δy.
$\Delta x = x_f - x_i$; velocity m/s; acceleration m/s²; time s.

Honors: add model conditions and a physical-root reminder.

4 / Keep the graph rules visible

- **x-t slope = velocity.** Horizontal line means at rest.
- **v-t slope = acceleration.** Horizontal line means constant velocity.
- **Signed v-t area = displacement.** Total distance adds the absolute areas.
- **a-t signed area = Δv.**
- Velocity and acceleration with the **same sign**: speeding up. Opposite signs: slowing down.

Write your sign convention

"Negative velocity" means motion in your negative direction. It does not automatically mean slowing down.

Check slope/area units against your interpretation.

5 / Build a projectile checklist

1. Sketch axes and choose a positive direction.
2. If θ is above horizontal:
   $v_{ix} = v_i \cos\theta$; $v_{iy} = v_i \sin\theta$.
3. Ignore air resistance: $a_x = 0$. With up positive, $a_y = -9.8$ m/s².
4. Use vertical motion to find the time for the target event; then use $\Delta x = v_{ix}t$.

Keep x/y KDEA lists separate. **Both axes share time, not velocity.** At the peak $v_y = 0$, but gravity still acts.

6 / Save space for error traps

- **Dropped:** $v_{iy} = 0$. **Horizontal launch:** $v_{iy} = 0$; v_{ix} may be nonzero.
- **At the peak:** only vertical velocity is zero; horizontal velocity can remain nonzero.
- Same launch/landing height? $\Delta y = 0$. Otherwise, use the actual height change.
- For angles in degrees, check **DEG** mode.
- Rearrange before substituting. Check units, direction, and whether the result fits the story.

Five-minute card test: Close your notes. Pick one fresh review problem. Find your first useful cue within 15 seconds, then solve with KDEA. If you stall, add the missing cue; remove a line you never use. **Explain every line before you bring the card.**